BIOINFORMATICS, GENOMICS AND BEYOND

Eric Winter M.S. | Winona State University | April 2026

"BIOINFORMATICS"
AN EXPANSIVE DEFINITION

- "the collection, classification, storage, and analysis of biochemical and biological information using computers especially as applied to molecular genetics and genomics" — Merriam-Webster
- From writing cloud-scale sequencing pipelines to building clinician-facing UX to curating variant knowledge bases
- One title, a dozen careers: the pipelines engineer, the clinical informatics developer, the data scientist, the visualization engineer…
- Plenty of opportunity to grow and evolve!

THIS TALK

- What is bioinformatics — and why it's broader than you think
- My work at Mayo Clinic
- Know your business — AI as a force multiplier
- The genome — scale, sequencing, and where the real work lives
- Clinical applications — interpretation, PRS, pharmacogenomics, cancer care
- Beyond genomics — transcriptomics, spatial, and the -omes
- Tools of the trade — pipelines, cloud, databases
- A day in the life — two careers, one field
- Getting started and career advice

MY WORK

- Lead Software Engineer and Architect — Mayo's Omics Data Platform
  - A data lake and genomics analytics platform for genomic research at scale on Google Cloud Platform.
- Research AI Services Lead — accelerating research discoveries with AI
- Clinical genomics laboratory support *(until recently)*
  - Pipelines, interpretation support software, genetics knowledge management.

Focused on the efficiency of those who analyze human variation — because the needs of the patient come first.

KNOW YOUR BUSINESS!

- Do NOT leave this to "someone else" to do for long. Dig in and invest.
- Engineers should:
  - Know their business
  - Learn the domain
  - Relate to Customers
  - Love your users
  - Build tools for a **PURPOSE**, something you're **EXCITED** about
- Not all engineers will — but the valuable ones will.
- **AI is your power tool** — domain knowledge is what makes you skilled enough to use it well
- Workers with advanced AI skills earn **56% more** than peers in the same roles — domain depth is what makes those skills stick — Dallas Fed, 2026

AI AND BIOINFORMATICS

- AI has already transformed the field — this is not a prediction
  - **AlphaFold 3** — protein structure and interaction prediction is largely solved
  - **Genomic foundation models** — Evo, Nucleotide Transformer, DNABERT-2, scGPT
  - **LLMs** are entering clinical variant interpretation workflows today
  - **Agentic tools** write and test substantial pipeline code
- What AI still can't do:
  - Tell you if a variant is *clinically significant* for *this patient*
  - Know whether a pipeline result makes *biological sense*
  - Understand why a genetic counselor needs *this* information in *this* workflow
  - Be accountable
- **That's you.** The bioinformatician who understands the science will direct AI — and be worth far more for it.

THE GENOME: SCALE, THEN SIMPLICITY

- **3 billion base pairs** — 500GB of raw data per sample at clinical coverage
  - 100 samples/day in a clinical lab = 50TB of raw data *per day*
- Sequencing that genome? **Getting easier and cheaper every year**
  - Panel (handful of genes) → WES (~20k genes) → WGS (everything)
  - Cost has dropped from $3B (Human Genome Project) to under $1,000 today
- Generating the sequence is the *easy* part
  - **What you do with it is where bioinformatics lives**

WHERE THE REAL WORK LIVES

- **Sequence analysis** — extracting the signal from the noise at massive scale
- **Quality assessment & control** — because a bad result can harm a patient
- **Interpretation** — turning a list of variants into an answer for a real person
- **Knowledge management** — so every patient benefits from every patient before them
- **Reporting** — communicating complex findings in a way that actually helps clinicians and patients
- **Discovery** — finding the genetic roots of disease, at scale, for the first time in history
- "*The $1,000 genome and the $100,000 analysis*" — this is where the value is. **This is where you come in.**

Schwarze et al. Genet Med 22, 85–94 (2020). https://doi.org/10.1038/s41436-019-0618-7

FROM CHEMISTRY TO DATA

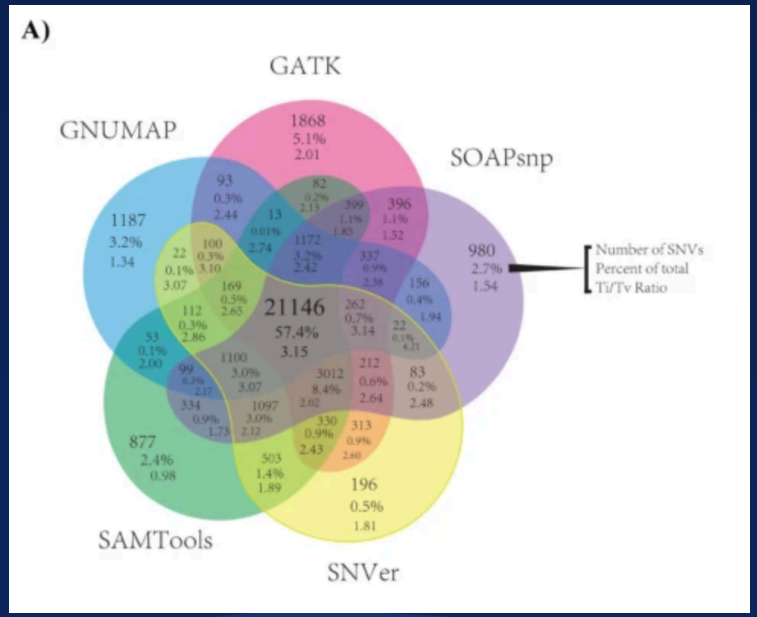
- A sequencer fragments DNA into billions of short reads (100–250 bp), each sequenced in parallel
  - The instrument produces billions of tiny images — software turns those into strings of A, T, C, G with quality scores
  - The industry is shifting toward long-read sequencing (PacBio) — better for complex regions and structural variants
- *Want to see how it actually works?* Illumina Sequencing Explained

THE ASSEMBLY PROBLEM

- Imagine shredding millions of copies of a book and reconstructing it from the fragments
- Two approaches:
  - **De-novo assembly** — compare every fragment against every other *(NP-hard territory — used when no reference exists)*
  - **Reference mapping** — align fragments against a known genome *(fast, but misses truly novel variation)*
- From aligned reads, the sample is compared position-by-position to the reference — differences are **variant calls**
- No two callers agree perfectly — the overlap between tools is the confidence zone

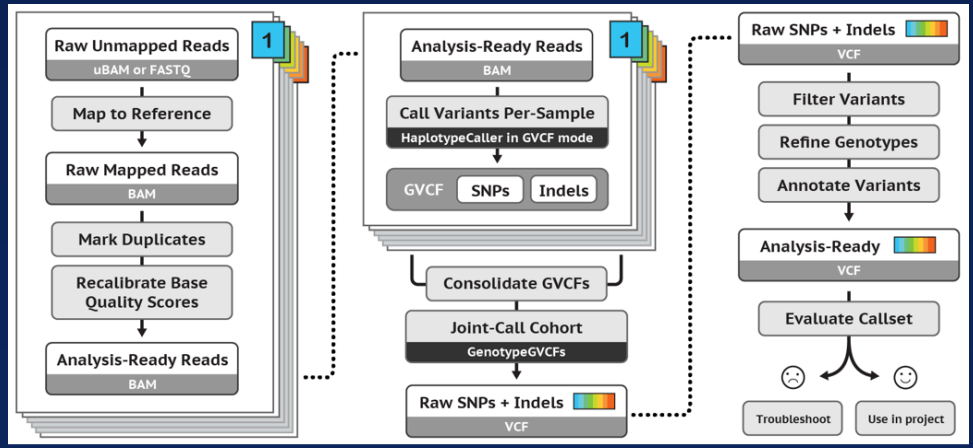
NO TWO CALLERS AGREE

- Run the same data through five different variant callers — they find **different variants**
- Only **57%** of called variants are found by all five tools
- The overlap is the confidence zone. The disagreement is where engineering judgment matters.
- Pipeline selection, tuning, and validation are real engineering and scientific problems



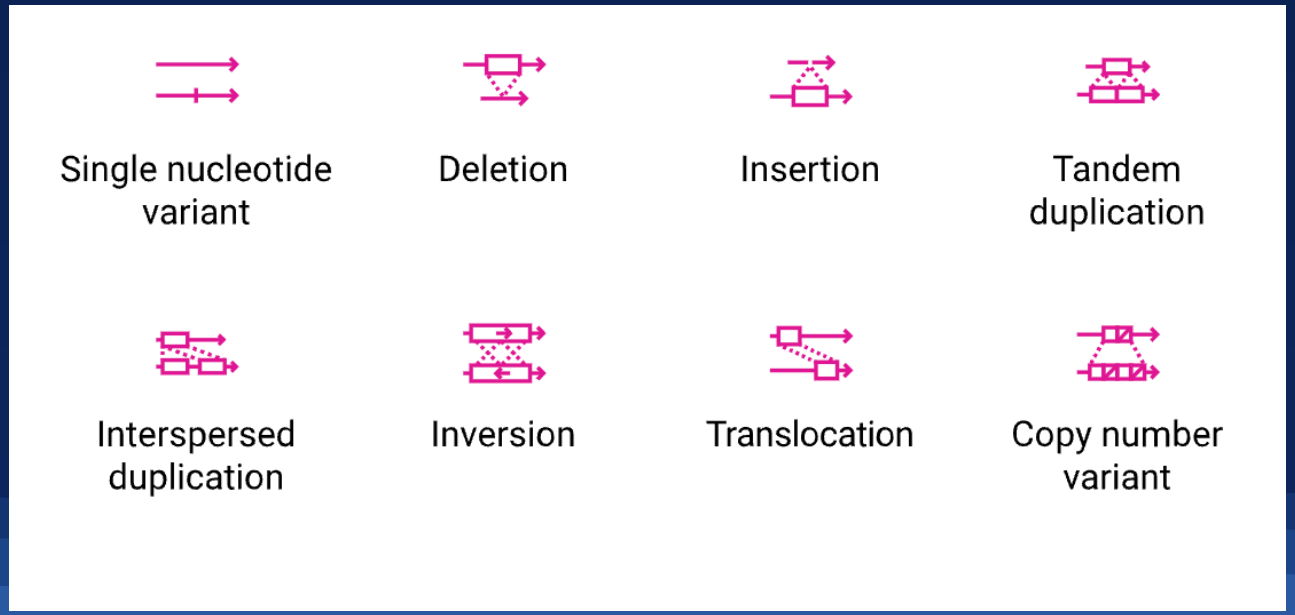
O'Rawe et al. Genome Medicine (2013).
https://genomemedicine.biomedcentral.com/articles/10.1186/gm432

THE INDUSTRY SETTLED ON THIS



Best Practices for SNP and Indel discovery in germline DNA - leveraging groundbreaking methods for combined power and scalability.

VARIANT TYPES



INTERPRETATION

- We now know all the variants the subject has from a reference … that is the easy part
- Often performed by genetic counselors reviewing literature
- 5 class system: Benign, Likely Benign, **Likely Pathogenic, Pathogenic**, Variant of Unknown Significance (VUS)
- AI is beginning to assist here — LLMs entering clinical variant interpretation workflows today

HOLY DATA!

- You have roughly 4 to 5 million SNPs from the human reference genome
- You have about 175 frameshift InDels
- 5% of your genome can be described as a 'structural variant'
- You have hundreds of 'Pathogenic' variants!

How do you tell the important stuff efficiently?

KNOWLEDGE MANAGEMENT

- Every variant interpreted once should never need to be interpreted from scratch again — the goal is to invest in new discovery, not re-work
- The funnel: **20k+ genes → ~2.5k tied to disease → ~80 actionable today**
- Jackson Labs CKB — one of the most important curated knowledge bases in clinical genomics. Explore it!

USER EXPERIENCE AND VISUALIZATION

- The greatest opportunity for improvement is here — data is growing exponentially, but the workforce isn't
- Only **~4.7k** certified genetic counselors and **~1.2k** medical geneticists in the U.S.
- Better tools don't just help — they're the only way to scale. **This is where engineers make the biggest difference.**

PREDICTING VARIANT IMPACT

- Algorithms that predict whether a mutation matters — using chemistry, evolutionary conservation, and machine learning
- **Missense prediction** (SIFT, PolyPhen) — does swapping this amino acid break the protein?
- **Splice site prediction** (SpliceAI, MaxEntScan) — does this mutation disrupt how genes are read?
- The frontier: deep learning models are increasingly outperforming classical tools

DATABASES THAT POWER IT ALL

- None of this works without massive, shared datasets — bioinformatics runs on open data
- **Reference databases** — dbSNP, 1000 Genomes, HGMD, dbNSFP, and ClinVar — the public archive linking variants to clinical significance
- **Population databases** — how common is this variant across different populations? A variant common in one group may be rare and significant in another
- gnomAD — the gold standard for population allele frequencies. Explore it!

GWAS AND BIOBANKING

- What if you could compare millions of genomes to find which variants are linked to disease? That's GWAS.
- The approach: test every variant position across an entire population against a phenotype (e.g. diabetes, heart disease) — a massive statistical sweep
- The catch: millions of simultaneous hypothesis tests means you need extreme significance thresholds ($p < 5\times10^{-8}$) to avoid false positives — a CS-scale multiple testing problem
- The key insight: the more genomes you bank with phenotype data, the more you can discover — **scale IS the science**
  - UK Biobank: 500k genomes. The discoveries accelerate as the dataset grows.
- A great primer: nature.com/articles/s43586-021-00056-9

POLYGENIC RISK SCORES (PRS)

- One variant = tiny effect. **Thousands combined = meaningful prediction.** This is PRS.
- How it's built: take GWAS effect sizes across thousands of variants, weight and combine them using ML — the result is a single score per person
- Clinical use today: risk stratification for heart disease, diabetes, breast cancer — helping doctors intervene **before** symptoms appear
- A CS problem through and through: feature selection, model training, cross-validation — all on population-scale genomic data
- The frontier: PRS is entering clinical workflows now, but challenges remain — scores built on European populations don't always transfer to other ancestries. **Fixing this is an open problem.**

PHARMACOGENOMICS

- Why does the same drug work for one patient and fail for another? Often the answer is in their genome.
- **Pharmacogenomics** uses genetic variants to predict drug response — right drug, right dose, right patient
- Already clinical today:
  - Warfarin dosing guided by CYP2C9/VKORC1 variants
  - Cancer therapies selected based on tumor mutations (e.g. EGFR inhibitors for specific lung cancers)
  - Antidepressant selection guided by CYP2D6 metabolizer status
- The engineering challenge: integrating genomic data into electronic health records so it's available **at the point of care**, not buried in a report
- This is personalized medicine — and it's one of the most immediate ways bioinformatics saves lives

EXAMPLE: GENOMICS IN CANCER CARE

- A patient is diagnosed with lung cancer. Here's where bioinformatics enters:
- **Tumor sequencing** — the tumor is biopsied and sequenced, looking for mutations driving the cancer
- **Variant interpretation** — is this mutation known to respond to a targeted therapy? Check ClinVar, CKB, OncoKB
- **Treatment selection** — EGFR mutation found → prescribe a targeted inhibitor instead of broad chemotherapy. Better outcomes, fewer side effects.
- **Monitoring** — liquid biopsy (cell-free DNA from a blood draw) can track whether the tumor is responding or developing resistance — without another biopsy
- **Every step of this pipeline was built by someone with your skills** — sequencing pipelines, variant callers, knowledge bases, clinical decision support, monitoring dashboards

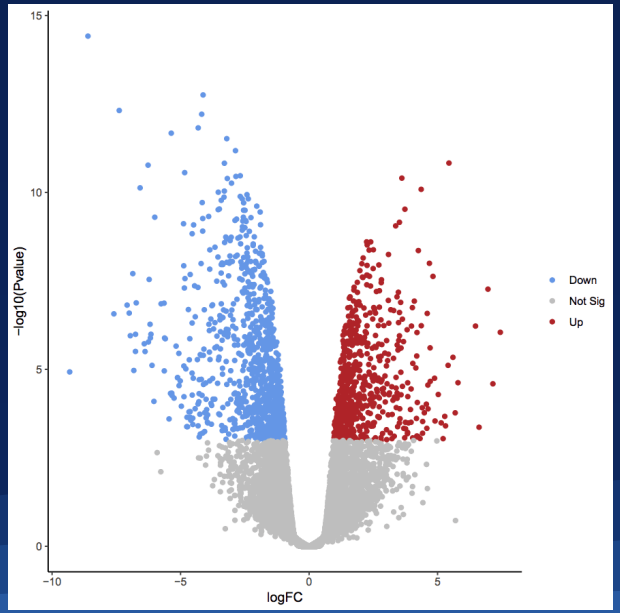
SO MANY MORE -OMES

- **Transcriptome** — What mRNA is floating around, meaning some level of expression?
- **Microbiome** — What microorganisms are influencing my phenotype/genotype?
- **Epigenome** — How is my DNA expression being up/down regulated by the environment?
- **Proteome** — What proteins are actually available in my cells? In what form?
- …Genomes are easy

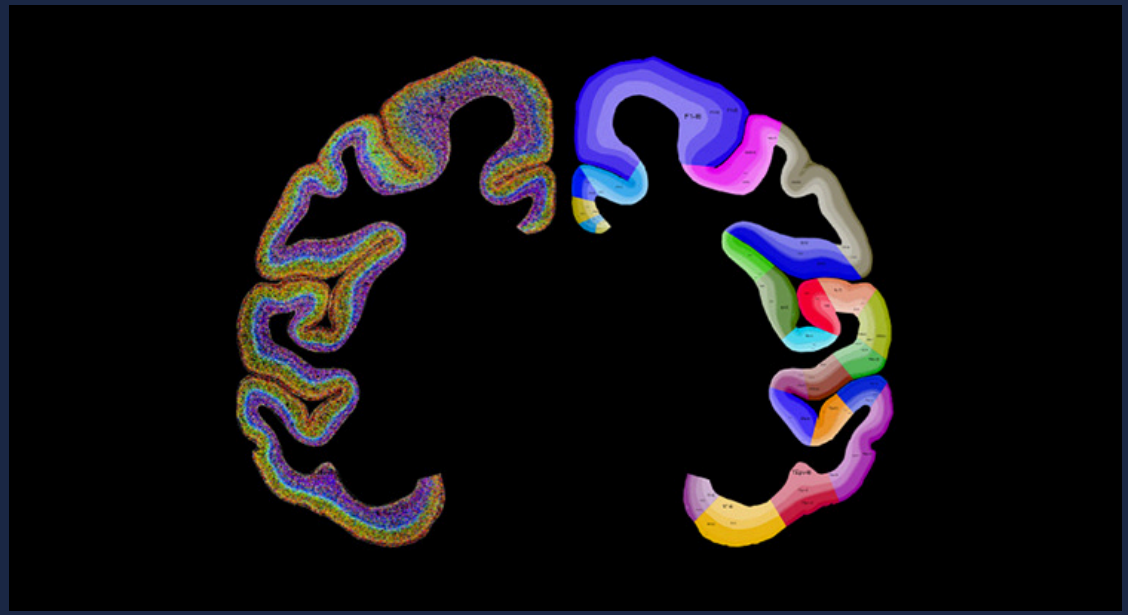
TRANSCRIPTOMICS

- DNA is just your genetic blueprint. RNA is your DNA responding to the environment and taking action.
- **RNA-Seq** identifies which genes are "turned on" — compare healthy vs. diseased cells to find which genes are misbehaving
- Why this matters as a research edge:
  - The genome is **static** — you're born with it. The transcriptome is **dynamic** — it changes with disease, treatment, time of day, even stress
  - Two patients with the same mutation can have completely different outcomes — transcriptomics helps explain why
  - Drug response: is the therapy actually changing gene expression? RNA-Seq can tell you
- Applications: disease profiling, drug response, developmental biology, single-cell analysis — millions of reads per sample
- Genomics tells you what *could* happen. Transcriptomics tells you what *is* happening.

GENE EXPRESSION VISUALIZATION



SPATIAL TRANSCRIPTOMICS



- See what genes are expressed and **where** — "Molecular Cartography" down to single cells on a tissue slide

WORKFLOW MANAGEMENT

- So much bioinformatics has to do with managing pipelines — reproducibility, parallelism, and portability matter enormously
- Manually assembling scripts is error-prone and doesn't scale
- Nextflow — the dominant workflow engine in bioinformatics today
  - nf-core — a community-curated collection of production-ready pipelines (including Sarek for GATK Best Practices)
  - Snakemake — another strong option, Python-based
- If you learn one tool for bioinformatics infrastructure, make it Nextflow

GENOMICS IN THE CLOUD

- It costs too much to bring the data to your science. **Your science needs to go to the data.**
- All major cloud providers (Google, AWS, Azure) have heavy investments in genomics infrastructure and public datasets
- The greatest benefits of genomics come at population scale — millions and billions of genomes. Only the cloud makes that possible.
- Get started: 1000 Genomes on GCP — real genomic data you can query today

LEARN THE BUSINESS

- Whatever domain you're in — read the textbooks your users read. You'll be a better engineer for it.
- Remember: AI is your power tool. **Domain knowledge is what makes you skilled enough to use it well.**
- Workers with domain depth + AI fluency earn **56% more** than peers without it
- The floor has risen — but so has the ceiling for those who invest

GETTING STARTED: THE SCIENCE

- Get your hands on real data — 1000 Genomes is free and waiting
- Learn the biology — pick one and go deep:
  - Genomic Data Science Specialization (Coursera)
  - Bioinformatics Specialization (Coursera)
  - Broad Institute courses — from the people who build GATK
- Read the textbooks your users read — even one molecular biology primer will change how you think about the problems

GETTING STARTED: THE ENGINEERING

- **Python** — the lingua franca of bioinformatics. Pandas, NumPy, and scikit-learn are your daily tools
- **R** — still strong for statistical genomics and visualization (ggplot2, Bioconductor)
- **Linux and Bash** — you'll live in the terminal. If you're not comfortable yet, get there fast
- **Nextflow** — the pipeline engine the field runs on (you've already seen why)
- **Parallel computing** (SLURM, cloud batch) — genomics workloads are embarrassingly parallel. Learn to exploit that

LEARN TO LEVERAGE A.I.

- You've seen where AI already is in bioinformatics — now make it part of your daily workflow
- **Use AI to write and test code** — GitHub Copilot, Claude, Cursor — these are standard tools, not shortcuts
- **Use AI to learn the domain faster** — ask it to explain papers, summarize variant databases, teach you biology
- **But never trust it blindly** — remember, only 1% of developers rely on AI alone. Your judgment is the filter.
- The engineers who thrive will combine **deep domain knowledge with AI fluency** — that's the whole talk in one line

A DAY IN THE LIFE: MAYA, GENOMIC BIOINFORMATICIAN

- **8:30 AM** — Reviews overnight variant interpretation queue. Flags two VUS cases for genetic counselor discussion.
- **10:00 AM** — Validates a new splice site prediction model against known pathogenic variants. Writes up results for the team.
- **1:00 PM** — Meets with genetic counselors to discuss a complex case — a patient with a rare variant not yet in ClinVar. Her domain knowledge shapes the interpretation.
- **3:00 PM** — Files a request to the engineering team: the reporting tool needs a new field for pharmacogenomic results. She writes the spec because she understands both the clinical need and the data model.
- **Maya's edge:** she reads the biology papers, attends tumor boards, and understands *why* the data matters — not just how to process it.

A DAY IN THE LIFE: ALEX, BIOINFORMATICS SOFTWARE ENGINEER

- **8:30 AM** — A Nextflow pipeline failed overnight on 200 samples. Debugs a memory issue in the variant calling step, pushes a fix, reruns.
- **10:00 AM** — Builds a new feature in the clinical reporting dashboard based on Maya's spec — adding pharmacogenomic annotations to patient reports.
- **1:00 PM** — Deploys an updated pipeline to GCP. Monitors throughput and cost per sample. Shaves 15% off runtime by tuning parallelism.
- **3:00 PM** — Pairs with a data scientist on a PRS model — helps optimize the training pipeline to run across the full biobank cohort.
- **Alex's edge:** he understands cloud infrastructure, pipeline engineering, and performance — but he also learned enough biology to ask the right questions and build what clinicians actually need.

YOUR CAREER STARTS NOW

- I started as a Research Technologist with degrees in Biology and Chemistry. I found I had little enthusiasm for running other people's experiments. Pivoting to Software Engineering was hard — and it was the best decision I ever made. **Be willing to pivot.**
- You're extremely lucky if you land your dream job right away — and that's okay. Start somewhere tolerable and grow from there.
- **Agility is your primary skill.** Small pivots, large pivots — the ability to redirect is what keeps you relevant. The job you have in five years probably doesn't exist yet.
- Build in public — get your code on GitHub. Contribute to open-source bioinformatics (nf-core, Bioconductor). Real résumé material, real community.
- Plug in: nf-core Slack, ASHG, AMIA, Bioinformatics Stack Exchange — your network will shape your career as much as your skills
- You're graduating into a field where a single engineer's work can change a patient's outcome. **Not many careers offer that.**

THANK YOU

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